CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

LAB MANUAL

Machining and Forming Processes

(ME 785)

ME – 785 MACHINING AND FORMING PROCESSES	
CO1	Student will learn Tool Geometry.
CO2	Student will learn various conventional machining processes.
CO3	Student will learn to Find Shear Angle.
CO4	Student will get the knowledge regarding Chip Thickness Analysis.
CO5	Students will understand and find the Cutting forces in orthogonal cutting.
CO6	Students get familiar to use Drilling Tool Dynamometer.
CO7	Student will learn Unconventional Forming.

List of Experiment (ME785- M&F)

Sr. No.	Title	Course Outcomes
1	Study of Tool Geometry	CO1
2	Study of various conventional machining processes	CO2
3	Experiment to Find Shear Angle	CO3
4	Chip Thickness Analysis	CO4
5	Cutting forces in orthogonal cutting	CO5
6	Experiment on a Drilling Tool Dynamometer	CO6
7	Study of Unconventional Forming	CO7

List of Assignment (ME785-M&F)

Sr. No.	Title	Course Outcomes
1	Assignment 1- Problems on machining and Tool life	CO3, CO5
2	Assignment 2- Questions on formed pipe defects and its remedies	CO7

INDEX

Sr. No.	Performance Date	Title	No. of Pages	Marks	Assessment Date	Sign
1		Study of Tool Geometry				
2		Study of various conventional machining processes				
3		Experiment to Find Shear Angle				
4		Chip Thickness Analysis				
5		Cutting forces in orthogonal cutting				
6		Experiment on a Drilling Tool Dynamometer				
7		Study of Unconventional Forming				

Machining and Forming Processes (ME-785) **Date:**

EXPERIMENT No. 1

STUDY OF TOOL GEOMETRY

AIM:

STUDY OF TOOL GEOMETRY OF SINGLE POINT, MULTI POINT CUTTING TOOL.

THEORY:

It is an instrumental media which helps to remove material from the work piece, and to give desired shape and size with the help of single point contact tool or multi point cutting tool. The relative motion is established between work and the cutting tool to undergo cutting process.

According to different methods-

- According to method of manufacturing-

  Forged tool, Tipped tool brazed to the carbon steel shank, Tipped tool fastened mechanically to the carbon steel shank.

- According to the method of holding the tool- Solid tool, Tool bit inserted in the tool holder.
- According to the method of using the tool- Turning, Chamfering, Thread cutting, Facing, Grooving, Forming, Boring, Internal thread cutting, parting off.
- According to the method of applying the feed- Right hand, Left hand, Round nose.

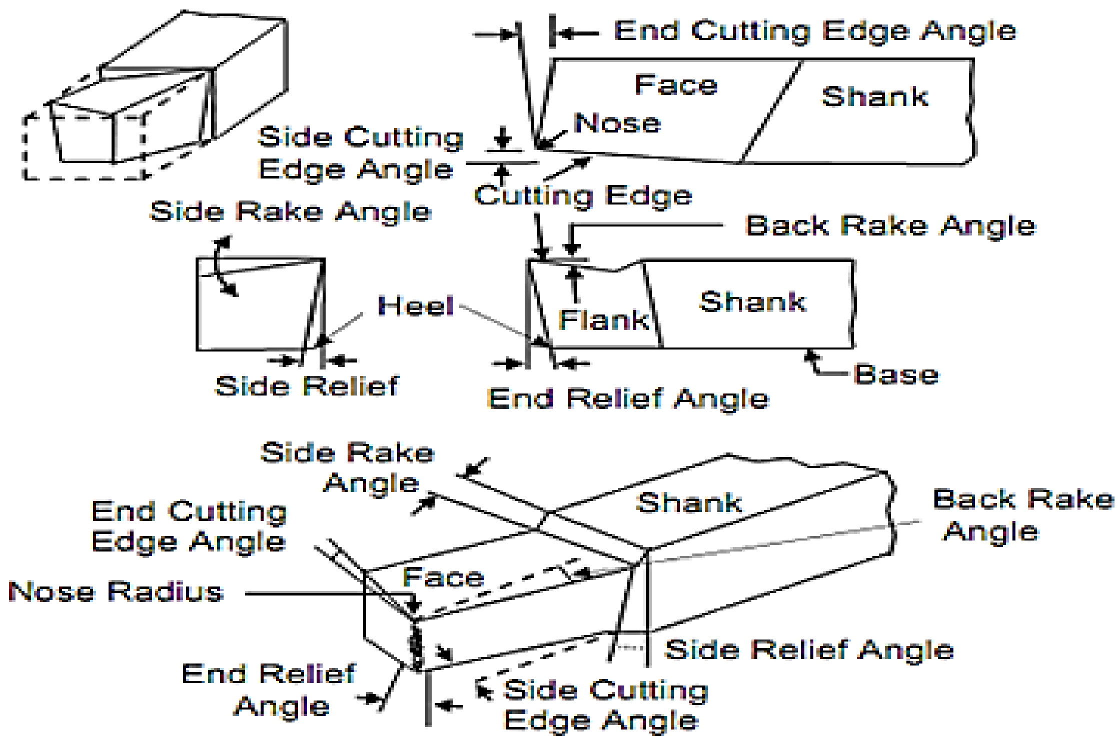
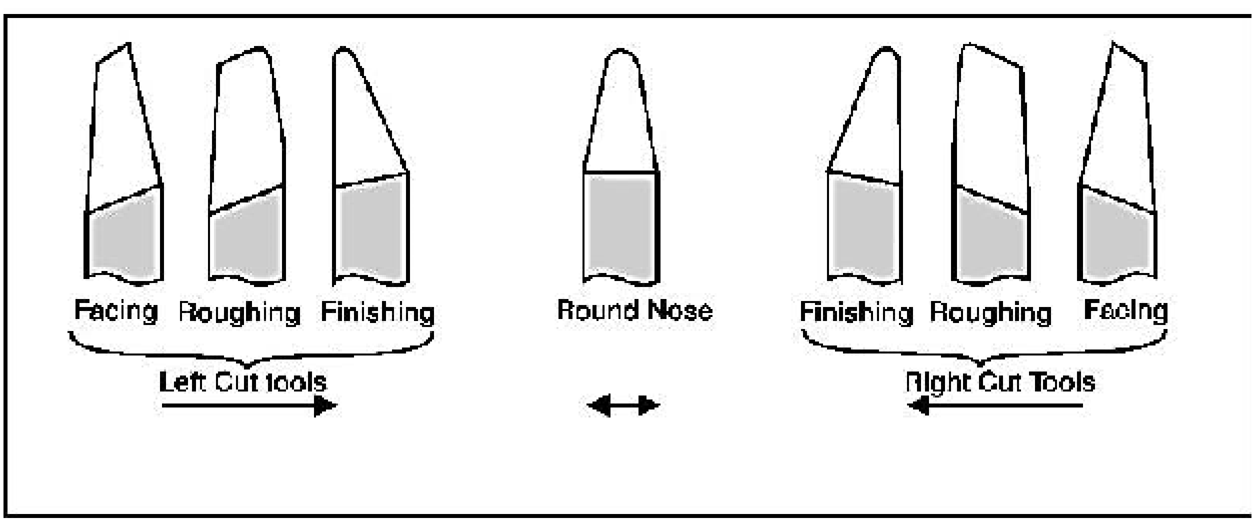
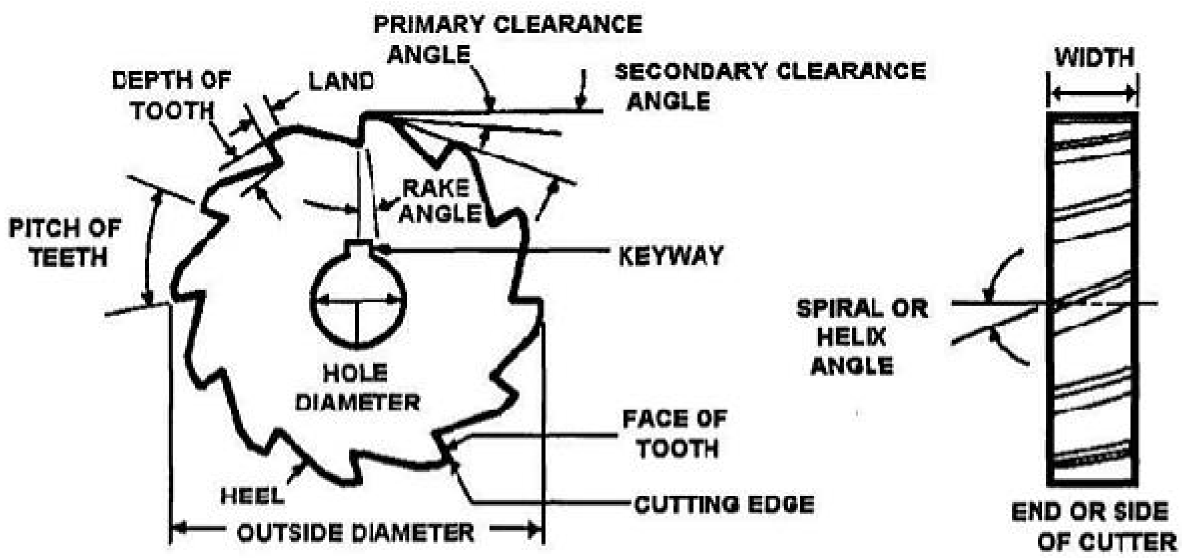


Fig. - Nomenclature of Single point cutting tool

SINGLE POINT CUTTING TOOL SHAPES



NOMENCLATURE OF MILLING CUTTER



QUESTIONS:

1. Explain with the help of neat sketch the complex geometry of a single point cutting tool and Multi point cutting tool.
2. State the different types of the tool nomenclature system. Explain briefly ASA system and ORS system of tool nomenclature with neat sketch.
3. Explain orthogonal cutting and oblique cutting.

Marks obtained	Signature of faculty	Date

Machining and Forming Processes (ME-785) **Date:**

EXPERIMENT No. 2

STUDY OF VARIOUS CONVENTIONAL MACHINING PROCESSES

AIM:

TO STUDY OF VARIOUS CONVENTIONAL MACHINING PROCESSES.

INTRODUCTION:

Machining is any of various processes in which a piece of raw material is cut into a desired final shape and size by a controlled material-removal process. The processes that have this common theme, controlled material removal, are today collectively known as subtractive manufacturing, in distinction from processes of controlled material addition, which are known as additive manufacturing. Exactly what the "controlled" part of the definition implies can vary, but it almost always implies the use of machine tools (in addition to just power tools and hand tools).

Machining is a part of the manufacture of many metal products, but it can also be used on materials such as wood, plastic, ceramic, and composites. A person who specializes in machining is called a machinist. A room, building, or company where machining is done is called a machine shop. Machining can be a business, a hobby, or both. Much of modern day machining is carried out by computer numerical control (CNC), in which computers are used to control the movement and operation of the mills, lathes, and other cutting machines.

"Traditional" or "conventional" machining processes, machine tools, such as lathes, milling machines, drill presses, or others, are used with a sharp cutting tool to remove material to achieve a desired geometry.

So ultimately Machining is any process in which a cutting tool is used to remove small chips of material from the workpiece (the workpiece is often called the "work"). To perform the operation, relative motion is required between the tool and the work. This relative motion is achieved in most machining operation by means of a primary motion, called "cutting speed" and a secondary motion called "feed". The shape of the tool and its penetration into the work surface, combined with these motions, produce the desired shape of the resulting work surface.

MACHINING OPERATIONS:

There are many kinds of machining operations, each of which is capable of generating a certain part geometry and surface texture.

In turning, a cutting tool with a single cutting edge is used to remove material from a rotating workpiece to generate a cylindrical shape. The primary motion is provided by rotating the workpiece, and the feed motion is achieved by moving the cutting tool slowly in a direction parallel to the axis of rotation of the workpiece.

Drilling is used to create a round hole. It is accomplished by a rotating tool that typically has two or four helical cutting edges. The tool is fed in a direction parallel to its axis of rotation into the workpiece to form the round hole.

In boring, a tool with a single bent pointed tip is advanced into a roughly made hole in a spinning workpiece to slightly enlarge the hole and improve its accuracy. It is a fine finishing operation used in the final stages of product manufacture.

Reaming is one of the sizing operations that remove a small amount of metal from a hole that already drilled.

In milling, a rotating tool with multiple cutting edges is moved slowly relative to the material to generate a plane or straight surface. The direction of the feed motion is perpendicular to the tool's axis of rotation. The speed motion is provided by the rotating milling cutter. The two basic forms of milling are peripheral milling and face milling.

Questions:

1. Discuss and attach any two research papers on “current research trends in conventional machining processes.”

Marks obtained	**Signature of faculty**	**Date**

Machining and Forming Processes (ME-785) **Date:**

EXPERIMENT No. 3

EXPERIMENT TO FIND SHEAR ANGLE

AIM:

TO EVALUATE SHEAR ANGLE AS A FUNCTION OF THE RAKE ANGLE OF THE TOOL

THEORY:

Orthogonal Cutting uses a wedge-shaped tool in which the cutting edge is perpendicular to the direction of cutting speed.

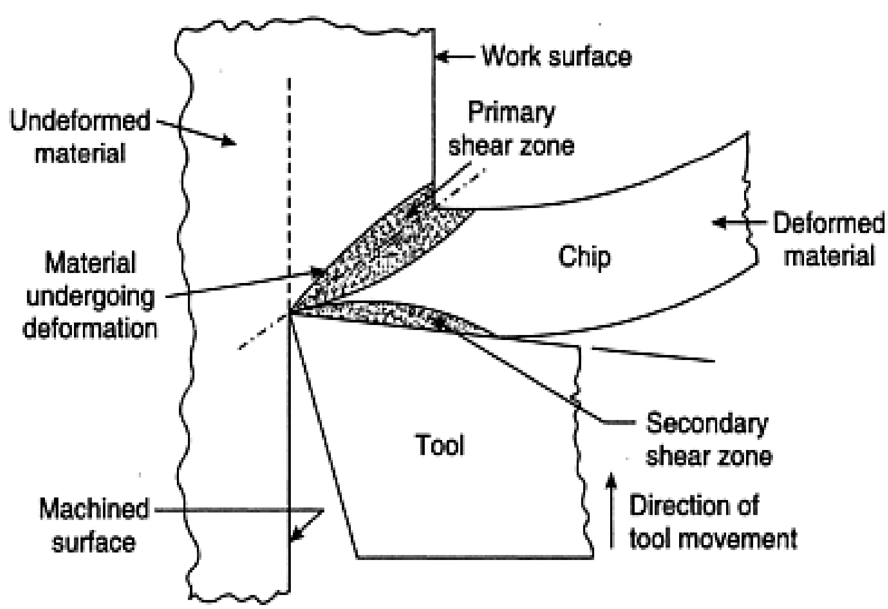


Fig 1: Deformation in metal cutting and primary and secondary shear zones

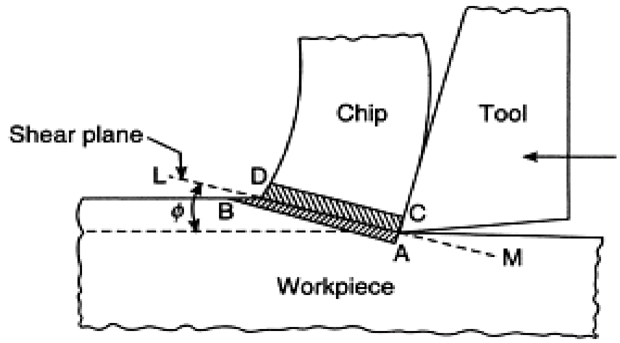


Fig 2: Illustrating the shear zone (ABCD), shear plane and shear angle (ϕ)

Primary deformation zone– shear in the work material
Secondary deformation zone – friction between chip and rake face
Tertiary deformation zone– friction between machined surface and flank face

Shear plane:

As the tool is forced into the material, the chip is formed by shear deformation along a plane called the shear plane.

Chip thickness ratio (or chip ratio):

$$r = \frac{t_o}{t_c}$$

Shear angle:

$$tan\emptyset = \frac{r\,cos\alpha}{1 - r\,sin\alpha}$$

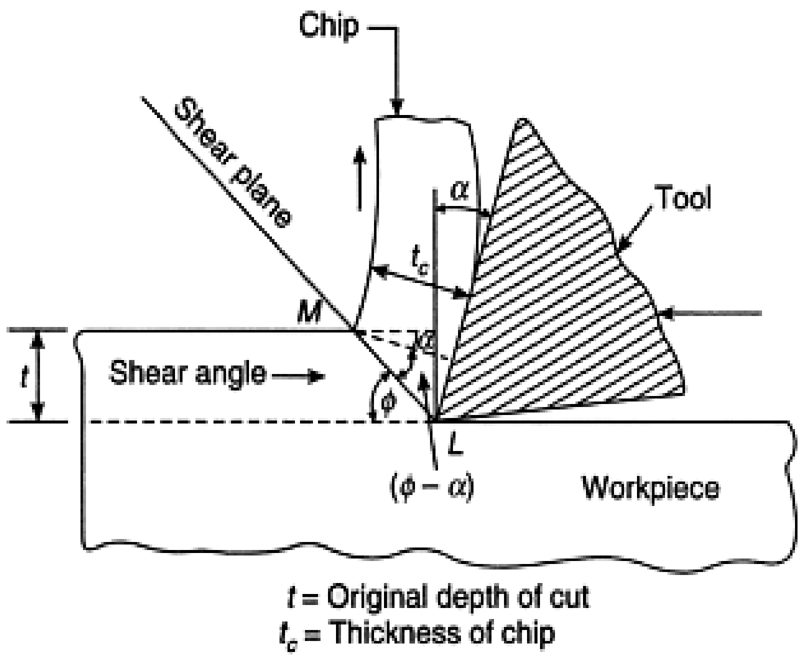


Fig 3: Illustrating shear angle (ϕ), shear plane and rake angle (α)

The Merchant equation defines the general relationship between rake angle, tool-chip friction, and shear plane angle as under:

$$\emptyset = 45 + \frac{\alpha}{2} - \frac{\beta}{2}$$

Where, $\emptyset$ = shear angle

α = rake angle

β = friction angle

OBSERVATIONS:

Tool Material:

Workpiece Material:

Spindle Speed (rpm):

Depth of Cut:

OBSERVATION TABLE:

Test	Side Rake Angle, α (Degrees)	Feed t_1, (mm)	Chip Thickness t_2 (mm)	Theoretical Shear Angle, ϕ(Degrees)	Practical Shear Angle ϕ(Degrees)
1	-5				
2	0				
3	+5				
4	+10				
5	+15				
6	+20				
7	+25				
8	+30				

GRAPH:

Rake angle vs. Chip thickness

Rake angle vs. Theoretical shear angle

Rake angle vs. Practical shear angle

CONCLUSION:

Marks obtained	Signature of faculty	Date

Machining and Forming Processes (ME-785) **Date:**

EXPERIMENT No. 4

CHIP THICKNESS ANALYSIS

AIM:

TO STUDY EFFECT OF CUTTING PARAMETERS ON CHIP THICKNESS

APPARATUS/INSTRUMENTS:
Lathe with a three-jaw chuck, Vernier Caliper, Ø20; 20 mm long M.S. rod.

PROCEDURE:
Set up the lathe and machine the MS rod for different set of cutting parameters. The cutting parameters are speed, feed and depth of cut. In the first set up vary the depth of cut, keeping the speed and feed constant. In the second set up vary the speed, keeping the feed and depth of cut constant. In the third setup, vary the feed, keeping the speed and depth of cut constant. Collect the chips and measure the thickness at four different points along the length

OBSERVATIONS:

1. Material: MS, Ø20 and 20 mm long

TABLE A: Speed =
Feed =

Depth of cut (mm)	Chip Thickness (mm)				
	T_1	T_2	T_3	T_4	Average

TABLE B: Feed =
Depth of cut =

Speed (rpm)	Chip Thickness (mm)				
	T_1	T_2	T_3	T_4	Average

TABLE C: Speed =
Depth of cut =

Feed (mm/rev)	Chip Thickness (mm)				
	T_1	T_2	T_3	T_4	Average

GRAPHS:

TABLE A: Depth of cut Vs Chip Thickness
TABLE B: Speed Vs Chip Thickness
TABLE C: Feed Vs Chip Thickness

CONCLUSION:

Marks obtained	Signature of faculty	Date

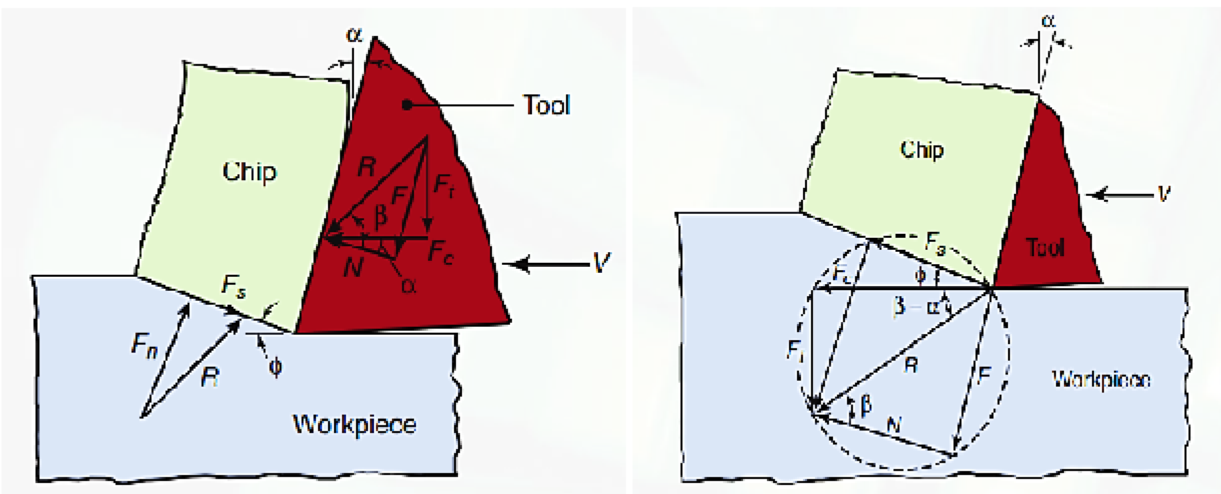
Machining and Forming Processes (ME-785) **Date:**

EXPERIMENT No. 5

CUTTING FORCES IN ORTHOGONAL CUTTING

AIM:
TO STUDY VARIOUS CUTTING FORCES IN ORTHOGONAL CUTTING.

INTRODUCTION:



Cutting force

$$F_c = R\cos(\beta - \alpha) = \frac{w t_o \tau \cos(\beta - \alpha)}{\sin\phi \cos(\phi + \beta - \alpha)}$$

Friction coefficient

$$\mu = \tan\beta = \frac{F_t + F_c \tan\alpha}{F_c - F_t \tan\alpha}$$

Forces acting on the tool in orthogonal cutting are shown in fig.

- Cutting force F_c acts in the direction of cutting speed V and supplies the energy required for the machining operation.
- Thrust force F_t acts in the direction normal to the cutting velocity.
- The two forces produces the resultant force R which can be resolved into two components acting on tool (friction force F resisting the flow of chip on rake face and normal force N perpendicular to F.)
- The resultant force R of F and N is oriented at β called the friction angle.

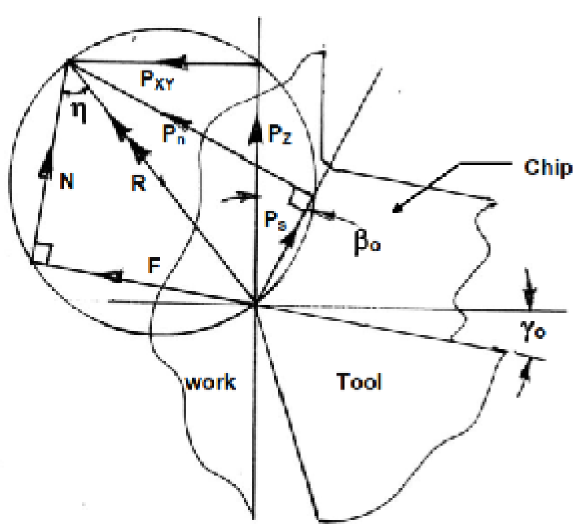
- F_s shear force that causes shear deformation and F_n normal to F_s.

Problem 1: During turning a ductile alloy by a tool of γ0 =100, it is found PZ=1000 N, Px=400 N, Py=300 N and ζ=2.5.Evaluate using MCD, the values of F,N and μ as well as PS and Pn for the above machining.

Problem 2: During turning a steel rod of diameter 160 mm at speed 560 rpm, feed 0.32 mm/rev and depth of cut 4.0 mm by a ceramic insert of geometry 00,-100, 60, 60, 150, 750, 0(mm)
The following was observed:
Pz=1600 N, Px=800 N and chip thickness=1mm. Determine with the help of MCD the possible values of F, N, μa, PS and Pn cutting power and specific energy consumption.

Problem 3: For turning a given steel rod by a tool of given geometry if shear force PS, Frictional force F and shear angle γ0 could be estimated to be 400N and 300N resp. then what would be the possible values of Px, PY And PZ? Use MCD.

Problem 4: During shaping like single point machining/turning) a steel plate at feed, 0.20 mm/stroke and depth 4 mm by a tool of λ = γ = 0o and φ = 90o Pz and Px were found (measured by dynamometer) to be 800 N and 400 N respectively, chip thickness, a2 is 0.4 mm. From the aforesaid conditions and using Merchant's Circle Diagram determine the yield shear strength of the work material in the machining condition.



Marks obtained	Signature of faculty	Date

Machining and Forming Processes (ME-785) **Date:**

EXPERIMENT No. 6

EXPERIMENT ON DRILLING TOOL DYNAMOMETER

AIM:

TO STUDY EFFECT OF INPUT PARAMETER ON FORCE IN DRILLING MACHINE.

APPARATUS:

Drilling machine
Drilling tool
Drilling tool dynamometer

THEORY:

In a drill there are two main cutting edges and a small chisel edge at the centre as shown in Fig.

The force components that develop during drilling operation are:

- a pair of tangential forces, P_{T1} and P_{T2} (equivalent to P_Z in turning) at the main cutting edges
- axial forces P_{X1} and P_{X2} acting in the same direction
- a pair of identical radial force components, P_{Y1} and P_{Y2}
- One additional axial force, P_{Xe} at the chisel edge which also removes material at the centre and under more stringent condition.

P_{T1} and P_{T2} produce the torque, T and causes power consumption P_C as,

$$T = P_T \times \tfrac{1}{2}(D)$$

and

$$P_C = 2\pi TN$$

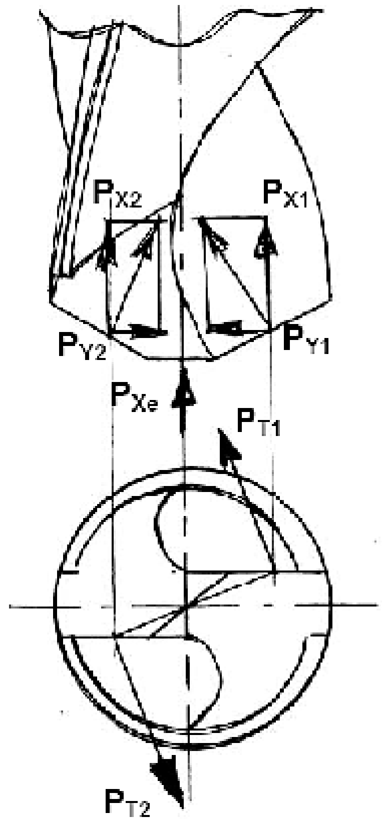
Where,

D = diameter of the drill
N = speed of the drill in rpm.

The total axial force P_{XT} which is normally very large in drilling, is provided by

$$P_{XT} = P_{X1} + P_{X2} + P_{Xe}$$

But there is no radial or transverse force as PY1 and PY2, being in opposite direction, nullify each other if the tool geometry is perfectly symmetrical.



OBSERVATION TABLE:
Drill Tool Material:
Drill Tool Diameter:

Sr. No	RPM	FEED	FORCE	TORQUE

Marks obtained	Signature of faculty	Date

Machining and Forming Processes (ME-785) **Date:**

EXPERIMENT No. 7

STUDY OF UNCONVENTIONAL FORMING

AIM:

TO STUDY THE DIFFERENT UNCONVENTIONAL FORMING PROCESSES.

INTRODUCTION:

Numerous technological forming processes achieve deformations, which are substantially higher than those achieved during the tensile test. One of the turbulently developing areas is the development of nano-structural materials, which is currently one of the priority areas of scientific research in the field of materials and forming technologies all over the world. This concern in particular forming of non-ferrous metals and their alloys. At the same time they bring significant reduction of manufacturing costs for products made from these materials. Importance of their use, particularly in the automotive industry, military and aerospace industries is ever increasing. Major world car makers, such as Opel, Audi, Jaguar, Ford, Fiat, Volvo and Toyota are at present developing an entirely new concept of fuel-efficient cars with a high share of aluminium and its alloys. Al alloys with ultra-fine structure serve as basic, initial semi-product. Their development uses technologies for achievement of nano-structural materials. Achievement of ultra-fine grained structure in initial material leads to substantial increase of plasticity and makes it possible to form materials in conditions of super plastic state.

Man’s quest to produce large, complicated components out of tougher materials has resulted in the search for new methods of manufacturer. The high energy rate forming processes makes use of the releases of the energy from high energy content substances such as explosive. The important processes under this category are explosive forming, electromagnetic forming and electro hydraulic forming.

Brief explanation of some of the unconventional forming processes is as under:

1. ECAP (Equal channel Angular Pressing):

Application of the ECAP method makes it possible to obtain an ultra-fined grain in larger volumes, when the initial cross-section does not change during extrusion. It finds an important of use namely in automotive industry and also in military and aerospace industries. The products manufactured by this technology meet the basic pre-requisites for their subsequent use at super-plastic forming. Diagram of the Equal Channel Angular Pressing (ECAP) method is shown in Fig.

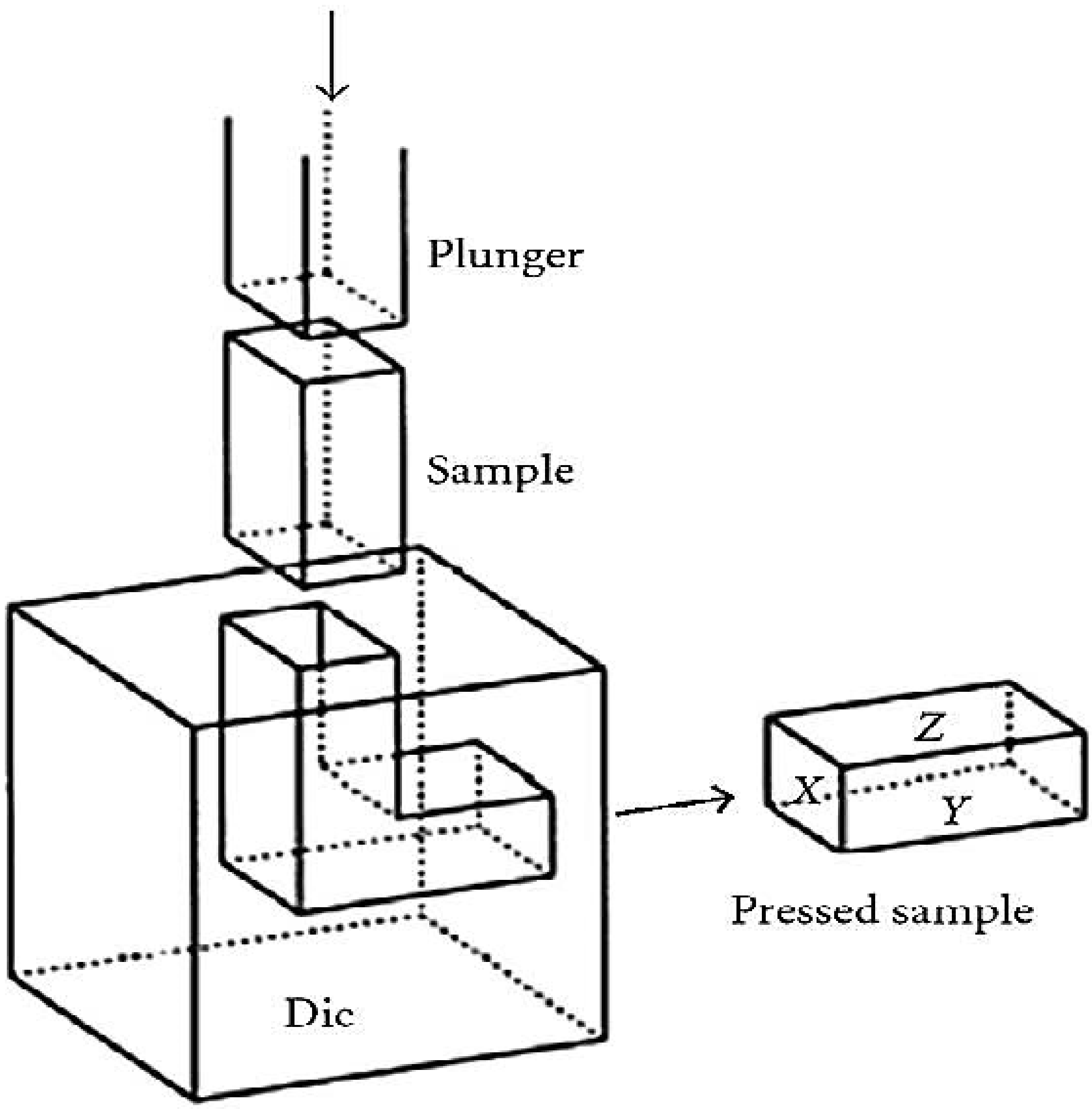


Fig.: Schematic illustration of the ECAP method

The machined specimen is inserted into the L-shaped matrix. For the case when the angle between two parts of the L-shaped matrix is equal to 90□, the test specimen is exposed to shear at the moment of transition from one part into another. It is evident that specimens are extruded in the channel without any change of their dimensions in the cross-section. By this the given process differs from majority of usual methods of metal forming, such as rolling and extrusion, which are accompanied by reduction of cross-section of the processed piece, and where the deformation is achieved by the change of the initial cross-section. In practice it is appropriate to define individual planes inside the specimen extruded by the ECAP technology, namely the plane X perpendicular to the longitudinal axis, and the planes Y and Z that are parallel to the lateral and top face of the specimen

2. Accumulative Roll Bonding:

The plate is in this process cut to two equally large parts, one part is thoroughly cleaned, the parts are put one onto another and then they are re-rolled to the original width. This rolling should ensure a diffusion welding of both plates, that's why this process requires mostly higher temperatures and low strain rates. The plate has after processing again the initial dimensions, it can be therefore cut again to two equally large parts and the process can be repeated as many times as needed. Schematic diagram of this method, which is currently intensively investigated in Japan, is shown in Fig. At present experiments are carried out attempting production of finely crystallised steel and aluminium alloy.

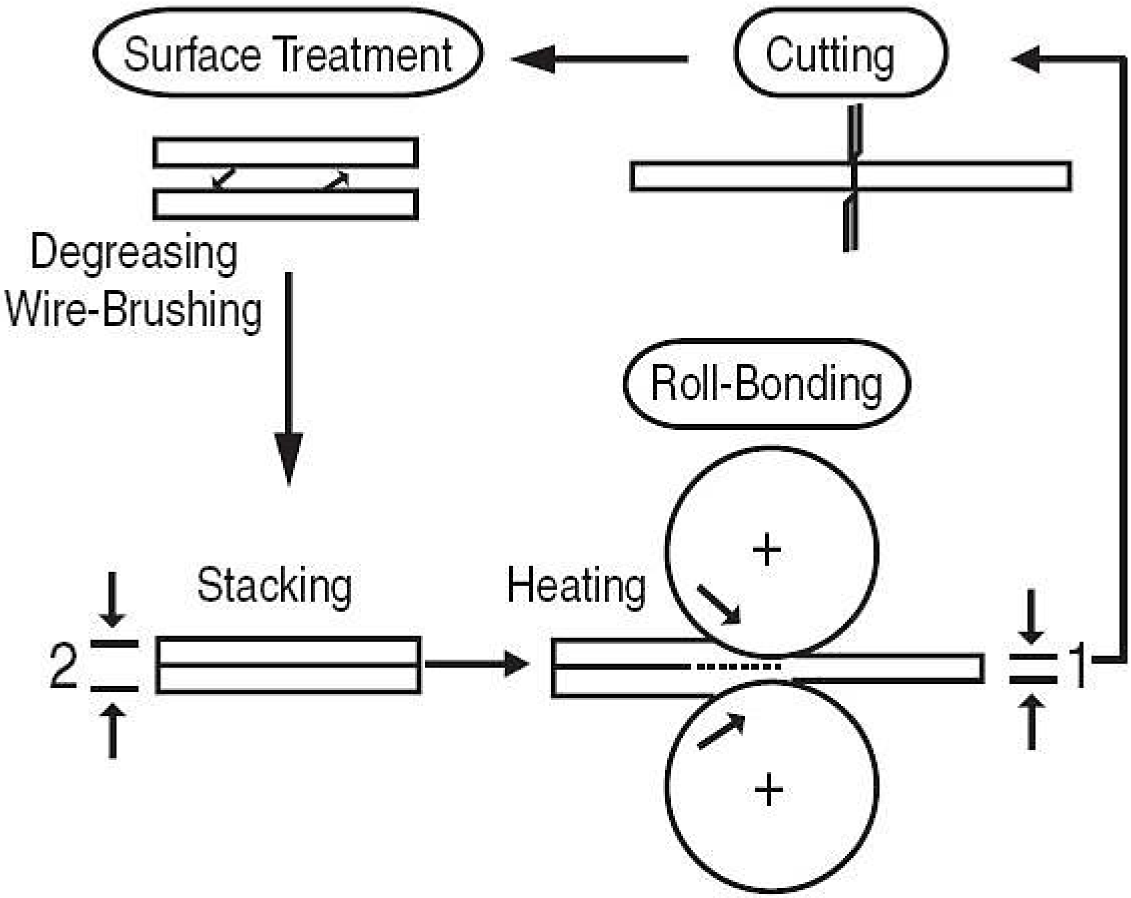


Fig.: Schematic illustration of the ARB method

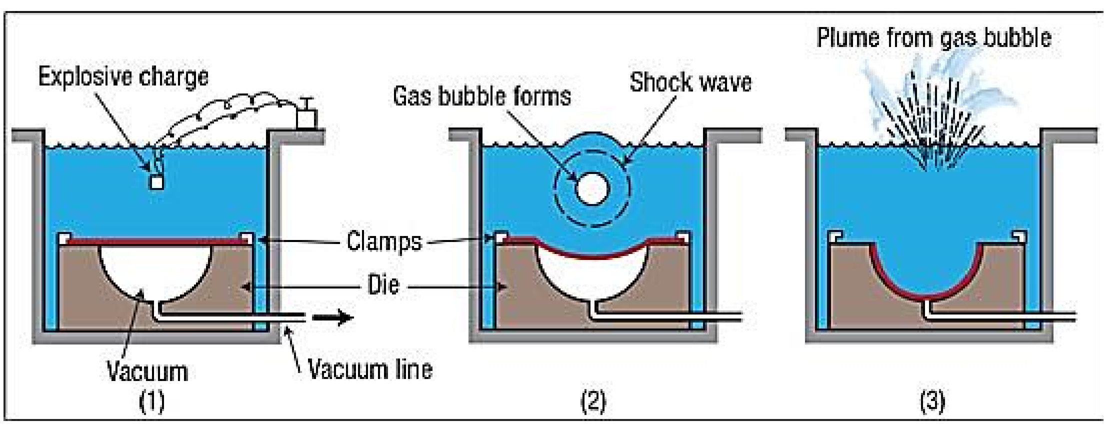
Disadvantage of this method is considerable heterogeneity of structure connected with an elongated shape of the deformed grain. Apart from this the contact surface of both plates must be carefully cleaned before each rolling, which increases production costs.

3. HIGH ENERGY RATE FORMING

High energy rate forming is the forming of sheet metal by a high energy surge, delivered over a very short time. Since the forming of the metal occurs so quickly, desirable materials for (HERF) will be ductile at high deformation speeds.

EXPLOSIVE FORMING-

Explosives can deliver a huge amount of power. Although most explosive detonations are destructive, the power from an explosive charge can be used to manufacture parts. An explosive forming process commonly used for the production of large parts is called a standoff system. Typically the mold and work piece are submerged in water. The sheet metal is secured over the mold by a ring clamp. Air is drawn out, creating a vacuum in the die cavity. An explosive is placed between the die cavity and the work, a certain distance from the work. This distance is called the standoff distance. Standoff distance depends on the size of the work, for larger parts it is usually about half the diameter of the blank. The explosive itself is also deeply submersed in water. Upon detonation, the shock wave travels through the water and delivers great energy to the work, forming it to the die cavity near instantaneously. This high energy rate forming process can be used to form big thick plates.

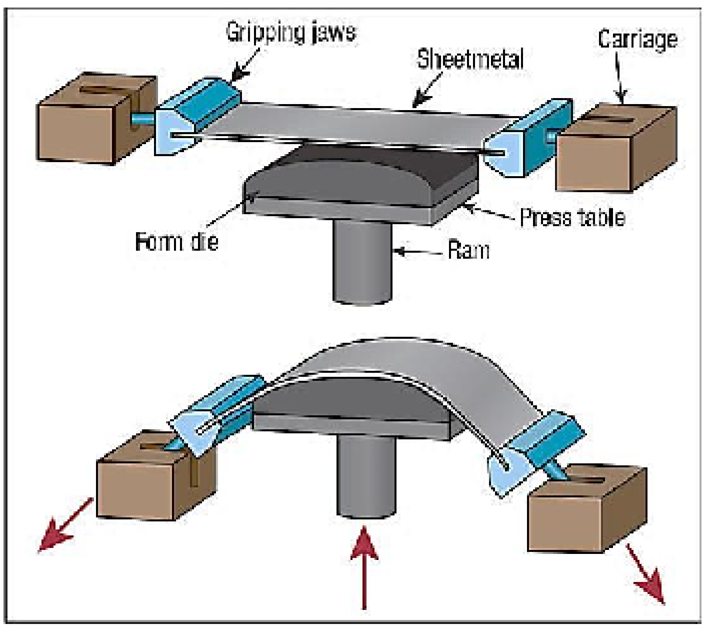


Explosive forming has a long cycle time and is suitable for low quantity production of large, unique parts. Mechanical properties imparted to the material as a result of the explosive forming process are similar to mechanical properties imparted to work manufactured by other forming processes. Molds can be made out of inexpensive or easy to shape materials, or molds can be made more permanent. Materials for molds include aluminum, wood, concrete, plastic, iron and steel. If a mold is manufactured from a material such as plastic, the low modulus of elasticity will greatly reduce spring back in the sheet metal, resulting in higher accuracy.

The amount of explosive depends upon the type of system used and the amount of pressure needed to form the part. The shock wave generated by the explosive travels along an expanding spherical front. Much of the energy from the shock wave is not absorbed by the work piece. A modified setup of the standoff system uses reflectors to focus the energy surge. This provides a more effective use of power and a smaller explosive can be used to form the same part. Another system called a confined system uses a canned explosive or cartridge. This is usually used for relatively smaller parts than the standoff system. All of the energy is directed into a closed container, the walls of which contain the die cavity. The energy from the canned explosive forces the sheet metal into the walls of the mold, forming the part. Safety is always a consideration when manufacturing by explosive forming, particularly with the confined system, where die failure is a significant concern.

4. **STRETCH FORMING**

Stretch forming simultaneously stretches and bends a piece of sheet metal to form a large contoured shape. Auto manufacturers use the process to produce outer-body panels; aircraft manufacturers stretch-form fuselage skin sections. Benefits of the process include a lack surface marring, distortions and ripples, and accurate alignment of complex profiles.



Stretch forming occurs in a stretch press, with the sheet metal securely held along its edges by gripping jaws. The gripping jaws, attached to a carriage, are pulled by pneumatic or hydraulic force to stretch the sheet (Fig.). The tooling used a stretch form block comprises a solid contoured piece against which the sheet metal is pressed. As the form die drives into the sheet, tensile forces increase until the sheet plastically deforms into its new shape.

Stampers select from two basic types of stretch-forming equipment: longitudinal and transverse. Longitudinal equipment stretches the workpiece along its length; transverse equipment stretches the workpiece along its width. Stretch presses are designed to be accurate and efficient, while CNC controls help provide part-to-part repeatability.

Questions:

1. Write down the differences between conventional and unconventional forming processes.
2. Write any two case studies on any unconventional forming processes.

Marks obtained	**Signature of faculty**	**Date**